

III. Final Rule 612 of Regulation NMS – Minimum Pricing Increment

Rule 612 of Regulation NMS establishes minimum pricing increments (also known as minimum price variations or tick sizes) for quotations and orders in NMS stocks. Specifically, preexisting Rule 612 stated that “[n]o national securities exchange, national securities association, alternative trading system, vendor, or broker or dealer shall display, rank, or accept from any person a bid or offer, an order, or an indication of interest in any NMS stock priced in an increment smaller than \$0.01 if that bid or offer, order, or indication of interest is priced equal to, or greater than, \$1.00 per share.”¹⁴⁵ Preexisting Rule 612(b) had similar language that applied to bids, offers, orders, and indications of interest in any NMS stock priced less than \$1.00 per share and specified that the minimum pricing increment could not be smaller than \$0.0001. Preexisting Rule 612 of Regulation NMS did not establish or include minimum pricing increments for transactions.¹⁴⁶

A. Issues Raised in the Existing Market Structure Related to Tick Sizes

The Proposing Release contains an extensive discussion of the development and the consideration by the Commission and market participants of Rule 612 since its adoption.¹⁴⁷ Since the adoption of Rule 612, there has been a marked increase in the trading volume of NMS stocks that would likely be priced with tighter spreads if their pricing was not constrained by the uniform \$0.01 minimum pricing increment required by preexisting Rule 612 for quotes and

¹⁴⁵ See 17 CFR 242.612.

¹⁴⁶ As discussed in the Proposing Release, the Commission granted exemptions from Rule 612 to various national securities exchanges’ retail liquidity programs (“RLPs”) as a way to allow them to compete with over-the-counter (“OTC”) market maker sub-penny price improvement. See Proposing Release, supra note 11, at 80271. Under the RLPs, exchanges can accept and rank certain quotes and orders from certain participants in sub-penny increments as small as \$0.001.

¹⁴⁷ See Proposing Release, supra note 11, at 80272-80273.

orders all NMS stocks priced equal to, or greater than, \$1.00 per share. Easing constraints on ticks for these NMS stocks will reduce transaction costs for market participants, including investors, and allow prices to be determined in a more competitive manner. In other words, the number and volume of NMS stocks that could benefit from the ability to quote in a minimum pricing increment that is smaller than \$0.01 (i.e., sub-pennies) has grown.

In the Proposing Release, the Commission considered data to evaluate and determine which NMS stocks, by number and by volume, would benefit from a reduced minimum pricing increment for quotes and orders that would allow for tighter spreads. While the Commission could not estimate the number of stocks that would have a TWAQS of \$0.008 or less due to the preexisting Rule 612 requirement that all orders priced equal to greater than \$1.00 per share have a \$0.01 minimum pricing increment, the Commission could estimate that 1,707 stocks, which represented approximately 64% of share volume and 37.9% of dollar volume in January through May 2022, had TWAQS that were less than \$0.016.¹⁴⁸ Additionally, 2,648 stocks, which represented approximately 17.9% of share volume and 22.3% of dollar volume in January through May 2022, traded with a spread that was greater than \$0.016 and less than or equal to \$0.04. More recently, the Commission analyzed NMS stocks in 2023 and identified 2,420 NMS stocks that had a TWAQS of \$0.015 or less; these NMS stocks represent about 74% of share volume and about 47% of dollar volume.¹⁴⁹

Prior to the Proposing Release, certain market participants conducted data analyses on the effects of Rule 612 and concluded that a \$0.01 minimum quoting increment may not be

¹⁴⁸ See Proposing Release, supra note 11, at 80280.

¹⁴⁹ See infra section VII.D.1.b, Table 3.

appropriate for all NMS stocks that are priced greater than or equal to \$1.00.¹⁵⁰ The Commission discussed these data analyses in the Proposing Release.¹⁵¹ One of these market participants, Cboe, submitted updated data analysis in two comment letters to the Proposing Release.¹⁵²

B. Proposal to Amend Rule 612

The Commission proposed variable minimum pricing increments for quotes and orders for NMS stocks priced at, or greater than, \$1.00 per share based on the TWAQS of a particular NMS stock. The Commission also proposed that the minimum pricing increment for executions be the same as, and correlate to, the minimum pricing increment for quoting on all trading venues (i.e., on-exchange and OTC), subject to certain exceptions.

Specifically, the Commission proposed that the minimum pricing increments for quotations, orders and executions in NMS stocks that are priced equal to or greater than \$1.00

¹⁵⁰ See, e.g., The Tick-Constrained Stock Problem by Phil Mackintosh (Jan. 20, 2022), [available at](http://www.nasdaq.com/articles/the-tick-constrained-stock-problem) <http://www.nasdaq.com/articles/the-tick-constrained-stock-problem>) (“Nasdaq Paper”). See also Petition for Rulemaking to Amend Rule 612 of Regulation NMS to Adopt Intelligent Tick-Size Regime, dated Dec. 16, 2019, submitted by John A. Zecca, Executive Vice President, Chief Legal Officer & Chief Regulatory Officer, Nasdaq Inc. [available at](https://www.sec.gov/rules/petitions/2019/petn4-756.pdf) <https://www.sec.gov/rules/petitions/2019/petn4-756.pdf> (“Nasdaq Intelligent Tick Proposal”); The Impact of Tick-constrained Securities on the U.S. Equity Market ([available at](http://www.nyse.com/publicdocs/Tick_Constrained_Stocks.pdf) http://www.nyse.com/publicdocs/Tick_Constrained_Stocks.pdf) (“NYSE White Paper”) (no date available); and Cboe Proposes Tick-Reduction Framework to Ensure Market Structure Benefits All Investors ([available at](https://www.cboe.com/insights/posts/cboe-proposes-tick-reduction-framework-to-ensure-market-structure-benefits-all-investors/) <https://www.cboe.com/insights/posts/cboe-proposes-tick-reduction-framework-to-ensure-market-structure-benefits-all-investors/>) (“Cboe Proposal”).

¹⁵¹ See Proposing Release, supra note 11, at 80274-80278.

¹⁵² See Letters from Angelo Evangelou, Cboe Global Markets, Inc., dated Feb. 28, 2023 (“Cboe Letter I”); Patrick Sexton, EVP, General Counsel & Corporate Secretary, Cboe Global Markets, Inc., dated Mar. 31, 2023 (“Cboe Letter II”) at Appendix A. See also Letter from Hope M. Jarkowski, General Counsel, NYSE Group, Inc., dated Mar. 27, 2023 (“NYSE Letter II”) (submitting for the record its paper entitled Price Improvement, tick harmonization & investor benefit (Aug. 22, 2022). This paper was described in the Proposing Release, supra note 11, at 80275; MEMX Letter, Appendix (submitting Tick-constrained Securities (Aug. 2021). This paper was described in the Proposing Release, supra note 11, at 80274. In the MEMX Letter, MEMX also submitted Tick-constrained Securities, The Tick Size Debate, Revisited (Jan. 2022) which analyzed a set of reverse splits on certain low-priced ProShares exchange-traded products (“ETPs”) and finding that the tick-constrained ETPs analyzed traded with significantly lower spreads post reverse split. This paper was described in the Proposing Release, supra note 11, at 80318.

per share would be variable and no smaller than: (1) \$0.001 if the TWAQS¹⁵³ for the NMS stock during the Evaluation Period¹⁵⁴ was equal to, or less than, \$0.008; (2) \$0.002, if the TWAQS for the NMS stock during the Evaluation Period was greater than \$0.008 but less than, or equal to \$0.016; (3) \$0.005, if the TWAQS for the NMS stock during the Evaluation Period was greater than \$0.016 but less than, or equal to, \$0.04; and (4) \$0.01 if the TWAQS for the NMS stock during the Evaluation Period was greater than \$0.04.¹⁵⁵ Further, as proposed, NMS stocks' TWAQS would have been measured quarterly based on one month of trading data.¹⁵⁶ In other words, it was proposed that the assignment of minimum pricing increments for the quoting and trading of NMS stocks priced equal to or greater than \$1.00 per share be done on a quarterly basis.

The Commission stated that it preliminarily believed that the proposed Rule 612 amendments would promote: (1) fair and orderly markets and economically efficient executions, particularly for tick-constrained NMS stocks and retail order flow; and (2) fair competition and equal regulation between OTC market makers, exchanges, and ATSS that compete for retail liquidity by requiring that NMS stocks trade with the same minimum pricing increment regardless of venue (i.e., on or off-exchange).¹⁵⁷ The Commission also stated that proposed Rule 612 would promote price discovery and price competition, particularly for tick-constrained stocks and retail order flow, by permitting the uniform quoting and trading of NMS stocks across

¹⁵³ See infra section III.C.7.b. See also proposed Rule 612(a).

¹⁵⁴ See infra section III.C.7.a. See also proposed Rule 612(a).

¹⁵⁵ See proposed Rule 612(c).

¹⁵⁶ See proposed Rule 612(a).

¹⁵⁷ See Proposing Release, supra note 11, at 80273 (discussing the competitive dynamic among exchanges, ATSS and OTC market makers).

trading venues, in finer increments, based on objective criteria. The Commission preliminarily believed that the proposed Rule 612 amendments would result in the pricing of quotes and orders being more in alignment with the principles of supply and demand.

C. Final Rule - Minimum Pricing Increments for Orders Priced Equal to or Greater than \$1.00 per Share

After considering comments, and analyzing additional data in response to those comments, the Commission is modifying and adopting the proposed amendments to Rule 612. As adopted, Rule 612(b)(2) provides that no national securities exchange, national securities association, ATS, vendor, or broker or dealer shall display, rank, or accept from any person a bid or offer, an order, or an indication of interest in any NMS stock in an increment smaller than required pursuant to either paragraph (i) or (ii) below if that bid or offer, order, or indication of interest is priced equal to or greater than \$1.00 per share:

(i) \$0.01, if the Time Weighted Average Quoted Spread for the NMS stock during the Evaluation Period was greater than, \$0.015; or

(ii) \$0.005, if the Time Weighted Average Quoted Spread for the NMS stock during the Evaluation Period was equal to or less than \$0.015.

Rule 612(b)(3) provides that no national securities exchange, national securities association, alternative trading system, vendor, or broker or dealer shall display, rank, or accept from any person a bid or offer, an order, or an indication of interest in any NMS stock priced in an increment smaller than \$0.0001 if that bid or offer, order, or indication of interest is priced less than \$1.00 per share.¹⁵⁸

¹⁵⁸ Rule 612(b)(3) is the same as preexisting Rule 612(b).

Further, as amended, minimum pricing increments for quotes and orders will be assigned on a semiannual basis using 3-months of trading data to calculate each NMS stock's TWAQS.¹⁵⁹ Therefore, as adopted, a minimum pricing increment of either \$0.01 or \$0.005 will be assigned to each NMS stock for quotes and orders that are priced equal to or greater than \$1.00 per share twice a year and will be operative for a six-month period.¹⁶⁰

The amendment differs from the proposal because rather than adding three proposed smaller minimum pricing increments for quotes and orders (\$0.005, \$0.002, \$0.001) to the current \$0.01 increment, only one additional minimum pricing increment (\$0.005) for NMS stocks that have a TWAQS of \$0.015 or less will be added. In addition, the amendment differs from the proposal as it (1) does not include a minimum pricing increment for trades, (2) modifies the Evaluation Period, and (3) provides for an implementation period.

1. General Comments and Discussion

The Commission received many comments on the proposal to amend Rule 612.¹⁶¹ Some commenters supported the need to amend Rule 612.¹⁶² Many individual commenters generally

¹⁵⁹ See Rule 612(a)(1).

¹⁶⁰ Some commenters suggested that the Commission consider wider quoting increments. See, e.g., Nasdaq Letter I; ASA Letter at 4; MEMX Letter at 20; Cboe, State Street, et al. Letter at 2; BIO Letter at 3; Invesco Letter at 3; Robinhood Letter at 39; Themis Letter at 5; Dimensional Letter at 2; and Letter from Tim Gately, Managing Director, Head of Equities Sales, Americas, Citigroup Global Markets, Inc., dated Mar. 31, 2023 ("Citigroup Letter") at 5. The Commission is not adopting a wider quoting increment for NMS stocks or a subset of NMS stocks as part of these amendments. As discussed throughout this release, the Commission is amending Rule 612 to address issues that developed related to the constraint that results from the \$0.01 minimum pricing increment. A wider quoting increment would not address these specific issues.

¹⁶¹ See *supra* note 82.

¹⁶² See, e.g., Form Letter Type A, of which 22 comments were received; Form Letter Type D, of which 255 comments were received; Form Letter Type G, of which 652 comments were received, [available at](https://www.sec.gov/comments/s7-30-22/s73022.htm) <https://www.sec.gov/comments/s7-30-22/s73022.htm>; IEX Letter I at 6; Letters from David Mechner, Chief Executive Officer, Pragma, LLC, dated Mar. 23, 2023 ("Pragma Letter"); Citigroup Letter at 4; MMI Letter at 3; Cboe, State Street, et al. Letter at 2; Nasdaq Letter I at 2; Managed Funds Letter dated March 30, 2023 at 11; letter from Joseph Scafidi, Global Head of Trading, and Carlos Oliveira, Head of Trading

supported the proposed amendments;¹⁶³ while some individual commenters agreed that Rule 612 should be amended but recommended that the proposal be modified.¹⁶⁴

Broadly, many commenters stated that preexisting Rule 612 should be amended in order to permit sub-penny quoting.¹⁶⁵ One commenter stated that for those stocks that are tick-constrained “[t]he one-cent increment for quoting can make it difficult for liquidity providers to fill orders and often results in higher trading costs.”¹⁶⁶ Another commenter stated that tick-constrained stocks experience wider quoted spreads, which results in “significantly increased transaction costs for investors,” and that these securities generally have longer queues and trade with “outsized notional liquidity at the NBBO.”¹⁶⁷ Several commenters stated that the “one-size-fits-all” requirement in Rule 612 should be revisited.¹⁶⁸ One commenter stated that Rule 612 impedes the ability of market participants to price some NMS stocks that would naturally be

Analytics and Market Structure, Brandes Investment Partners, L.P., dated Mar. 23, 2023 (endorsed by Adam Conn, Director, Baillie Gifford (Overseas) Ltd. et al.) (“Brandes Letter”) at 1; Angel Letter at 5; TradeStation Letter; Vanguard Letter at 4; B. Riley Letter at 1; JPMorgan Letter at 4; and UBS Letter at 10.

¹⁶³ See, e.g., Form Letter Type D, of which 255 comments were received; Form Letter Type E, of which 14 comments were received; and Form Letter Type G, of which 652 comments were received, available at <https://www.sec.gov/comments/s7-30-22/s73022.htm>; Letter from Bibambop RIP, dated Mar. 16, 2023; Letter from Binh Tran, dated Mar. 4, 2023; Letter from Jerry Pang, dated Mar. 4, 2023; Letter from Charlie Chen, dated Mar. 1, 2023; Letter from Daniel Song, dated Jan. 12, 2023; Letter from Deok Park, dated Dec. 26, 2023; and Letter from Clarissa West, dated Apr. 1, 2023.

¹⁶⁴ See, e.g., Form Letter Type H, of which 853 comments were received, available at <https://www.sec.gov/comments/s7-30-22/s73022.htm>.

¹⁶⁵ See, e.g., Letter from Stephen W. Hall, Legal Director and Securities Specialist, Better Markets, Inc., dated Oct. 31, 2023 (“Better Markets Letter II”) at 3; SIFMA Letter II; Brandes Letter at 1; ICI Letter I; BlackRock Letter; B. Riley Securities Letter; JPMorgan Letter at 4; Cambridge Letter at 6; Invesco Letter at 3; UBS Letter at 10; Citigroup Letter at 4; TradeStation Letter at 6; letters from individuals, including the Form Letter Type D, of which 255 comments were received; Form Letter Type G, of which 652 comments were received; and Form Letter Type H, of which 853 comments were received, available at <https://www.sec.gov/comments/s7-30-22/s73022.htm>.

¹⁶⁶ See ASA Letter at 4.

¹⁶⁷ See MEMX Letter at 9.

¹⁶⁸ See, e.g., SIFMA Letter II at 33; BlackRock Letter at 5; Citigroup Letter at 4; and MMI Letter at 5; UBS Letter at 10; Letter from Lawrence Harris, Ph.D, CFA, Professor of Finance and Business Economics, USC Marshall School of Business, dated Dec. 18, 2023 (“Harris Letter”) at 8.

priced within the penny spread.¹⁶⁹ The adopted minimum quoting increment of \$0.005 will enable the targeted NMS stocks to be more naturally priced based on the principles of supply and demand within the penny spread.

Generally, comments from individuals supported the proposal without any additional suggested changes.¹⁷⁰ One commenter stated of the proposal, “[t]his means that the pricing of stocks will be more precise and accurate, ensuring that I can get the best possible price for my trades.”¹⁷¹ Another commenter stated that “[a]llowing for sub-penny pricing will enable buyers to obtain lower prices from willing sellers and sellers to obtain higher prices from willing buyers, resulting in a more efficient market.”¹⁷² Comments from other market participants, including exchanges,¹⁷³ broker-dealers, and institutional investors¹⁷⁴ recommended modifying the proposal

¹⁶⁹ See Better Markets Letter II at 8.

¹⁷⁰ See, e.g., Form Letter Type A, of which 22 comments were received ; Form Letter Type D, of which 255 comments were received; Form Letter Type E, of which 14 comments were received; Form Letter Type G, of which 652 comments were received; Form Letter Type I, of which 22 comments were received; Form Letter Type J, of which 15 comments were received; and Form Letter Type K, of which 22 comments were received, available at <https://www.sec.gov/comments/s7-30-22/s73022.htm>.

¹⁷¹ Letter from John dated Feb. 23, 2023.

¹⁷² Letter from Nevin Varghese dated Dec. 26, 2022.

¹⁷³ See IEX Letter I at 6; Cboe, State Street, et al. Letter at 2; Nasdaq Letter I at 14; MEMX Letter at 18; and Cboe Letter II at 3.

¹⁷⁴ See Capital Group Letter at 4; ICI Letter I at 5-6; Vanguard Letter at 4-5; Invesco Letter at 3; Schwab Letter II at 6; T. Rowe Price Letter at 4; Fidelity Letter at 14; Brandes Investment Letter dated March 31, 2023 at 2; Ontario Teachers, Alberta Investment, CalSTRS, CalPERS, Canada Pension, and Texas Retirement Letter dated Mar. 31, 2023 at 2 (“Ontario Teachers et al. Letter”); BlackRock Letter at 5; Dimensional Letter at 2; B. Riley Letter at 1; and Letter from Christopher P. Bowker Jr., Director of Global Equity Trading, Boston Partners Global Investors, Inc., Joe Mariano, Senior Vice President, Global Head of Trading, Calamos Advisors LLC, Melissa F. Hinmon, Director of Equity Trading, Glenmede Investment Management, Dan Royal, Global Head of Equity Trading, Janus Henderson Investors US LLC, dated Apr. 6, 2023 (“Boston Partners, Calamos Advisors, Glenmede Investment, and Janus Henderson Letter”); State Street Letter at 3; NYSE, Schwab, and Citadel Letter at 2; Letter from John Zhu, Head of Trading, Optiver US LLC, dated Mar. 15, 2023 (“Optiver Letter”) at 4; Pragma Letter at 1; Cboe, State Street, et al. Letter at 2; Letter from Milan Galik, Chief Executive Officer, Interactive Brokers Group, Interactive Brokers LLC, dated Mar. 30, 2023 (“Interactive Brokers Letter”) at 5; RBC Letter at 3; Morgan Stanley Letter at 3-4; JPMorgan Letter at 4-5; Letter from at 2; Joe Wald, Managing Director & Co-Head of Electronic Trading, Eric Stockland, Managing Director, Global Markets, Brad A. Rothbaum, Managing Director & Head U.S. Global Markets, Chief Operating Officer & Head of the U.S. Branches, and Michael Forlenza, Managing

to Rule 612 to reduce the number of potential minimum quoting increments. Some commenters stated that further reduction of the minimum pricing increment for quotes and orders may be warranted for certain NMS stocks “in the future” but that a \$0.005 increment should be implemented and studied before any further reductions.¹⁷⁵ For the reasons discussed throughout, in response to commenters, the Commission is adopting amended Rule 612. Compared to the initial proposal, the modified amendments will be easier for market participants to implement and adapt to.

One commenter suggested that the Commission use its exemptive authority to reduce minimum pricing increments and access fees in a manner similar to that requested by MEMX.¹⁷⁶ MEMX requested an increment of \$0.005 for NMS stocks that are “tick-constrained” (defined by MEMX as stocks that trade with an average quoted spread of \$0.011 or less).¹⁷⁷ The commenter recommended this course of action as a means to gather data on sub-penny pricing increments to help determine whether, and to what degree, the proposed modifications were warranted.¹⁷⁸ The commenter also stated that using an exemption to test a reduction of minimum pricing increments and the access fee caps could include an expiration and a “roll-back” plan should unintended consequences become apparent.¹⁷⁹ Other commenters recommended that the Commission reduce the minimum pricing increments for a sample of stocks so that data could be

Director & Head of U.S. Capital Markets Compliance, BMO Capital Markets Corp., dated Mar. 31, 2023 (“BMO Letter”); Brandes Investment Letter dated March 23, 2023 at 2; B Riley Letter at 1; Themis Letter ; UBS Letter at 10; Citigroup Global Letter at 4-5; and Jefferies Letter at 3.

¹⁷⁵ See, e.g., BlackRock Letter at 6 and B. Riley Letter at 1.

¹⁷⁶ See Jefferies Letter. See also Proposing Release, supra note 11, at 80277 for a discussion of the MEMX request for exemption.

¹⁷⁷ See Proposing Release, supra note 11, at 80277 for a discussion of the MEMX request for exemption.

¹⁷⁸ See Jefferies Letter at 2.

¹⁷⁹ Id. at 4.

gathered and evaluated before changes were adopted on a more widespread basis.¹⁸⁰ Finally, one commenter recommended that the Commission establish a “transparent structured process to evaluate whether proposed changes to minimum pricing increments and access fees are actually improving the execution experience” and that a “clearly articulated off-ramp/kill-switch to unwind these changes” be in place to return to current minimum pricing increments and the access fee caps.¹⁸¹ Another commenter stated that if the Commission adopted a modified amendment to Rule 612 that such modification should be re-proposed for public comment.¹⁸²

An exemption, other temporary course of action, such as a pilot or sample reduction, or a re-proposal of the adopted amendments is not warranted. The Commission and market participants already have provided data and analyses that support amending Rule 612 to address tick constraints.¹⁸³ As discussed throughout this release, the adopted amendments to Rule 612 will allow NMS stocks that are experiencing tick constraints with the \$0.01 minimum pricing increment to be priced more competitively (i.e., reduce quoted spreads) and reduce transaction costs for liquidity demanders. The amendments to minimum pricing increments are designed to appropriately address significant concerns related to Rule 612.¹⁸⁴ One of the primary goals of the proposal and the adopted amendments is to alleviate tick constraints.

Reducing the minimum quoting increment for quotes and orders to \$0.005 for certain NMS stocks will enable such stocks to quote with tighter spreads, which in return reduces the

¹⁸⁰ See, e.g., Cboe, State Street, et al. Letter at 2; letter from Carlo Passeri, Vice President Biotechnology Innovation Organization (“BIO Letter”), dated Mar. 30, 2023; and State Street Letter at 3; MMI Letter at 3-7.

¹⁸¹ See Citigroup Letter at 6. With regard to the comment about an “off-ramp/kill-switch,” should the Commission observe trends detrimental to investors, the Commission could take appropriate action.

¹⁸² See Citadel Letter II at 3.

¹⁸³ See, e.g., MEMX Letter, Pragma Letter; IEX Letter I; and Nasdaq Letter I. See infra section VII.D.1.b.

¹⁸⁴ See supra section III.A.